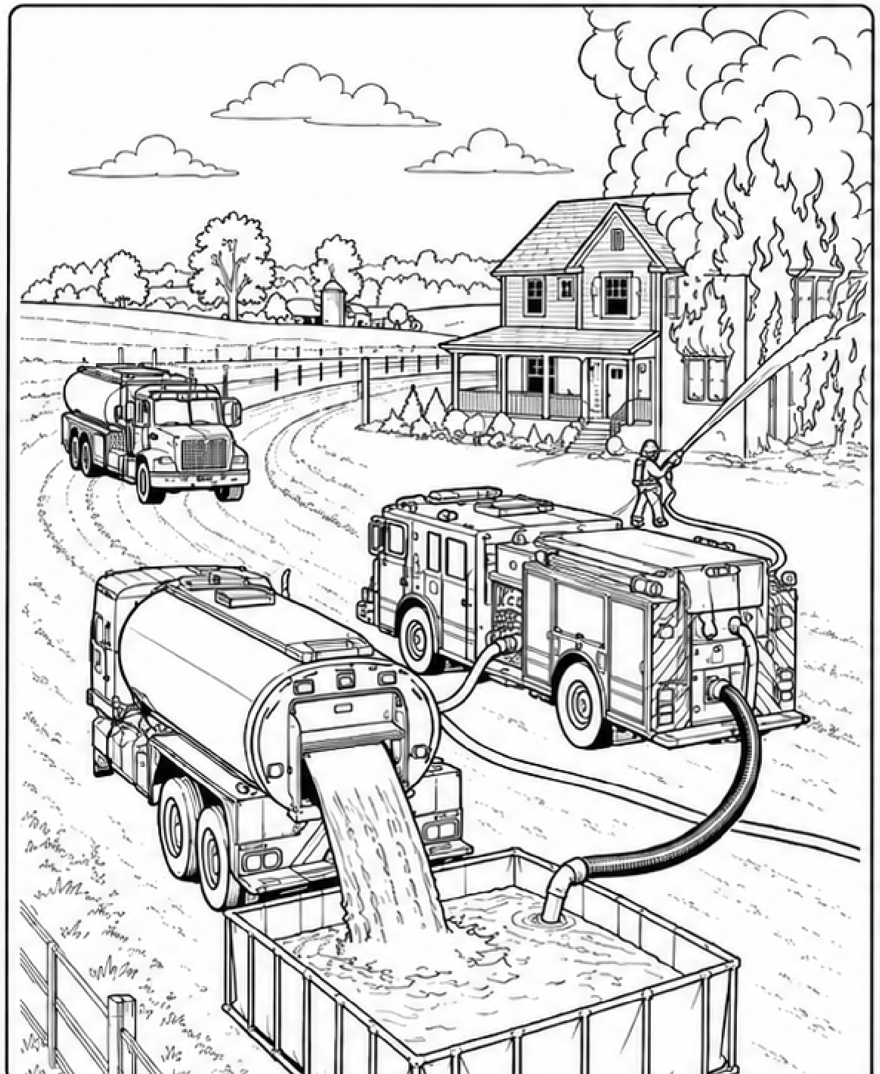
FirePumpSim Printable Scenario

Driver/Operator Practice Worksheet

Student Name: ____________________ **Date:** ______________ **Class:** ______________ **Score:** ________

Scenario: Rural Water Supply - Tanker Shuttle House Fire



Given / Reference

- **Apparatus:** Engine 181 drafting from portable tank
- **Water source:** Portable dump tank supplied by water tender shuttle
- **Portable tank starting water:** 2,000 gallons
- **Attack line:** 200 ft of 1¾-inch hose
- **Nozzle:** Fog nozzle, 150 GPM at 100 PSI
- **Hose coefficient (C):** 15.5
- **Appliance loss:** 0 PSI
- **Elevation change:** 0 PSI

Student Questions

1. Using $FL = C \times Q^2 \times L$, what is the friction loss in the 200-foot 1¾-inch attack line?

   ____

2. What pump discharge pressure (PDP) is required to supply the current attack line?

   ____

3. A second line uses 150 ft of 2½-inch hose (C = 2) with a fog nozzle flowing 200 GPM at 100 PSI. What pump discharge pressure (PDP) is required for that line?

   ____

4. If only one 150-GPM attack line is flowing, approximately how many minutes of water are available from the 2,000-gallon portable tank?

   ____

5. If the original 150-GPM line and the second 200-GPM line from Question 3 both flow for 5 minutes before the next tender arrives, how many gallons would remain in the portable tank?

   ____

Required Equations

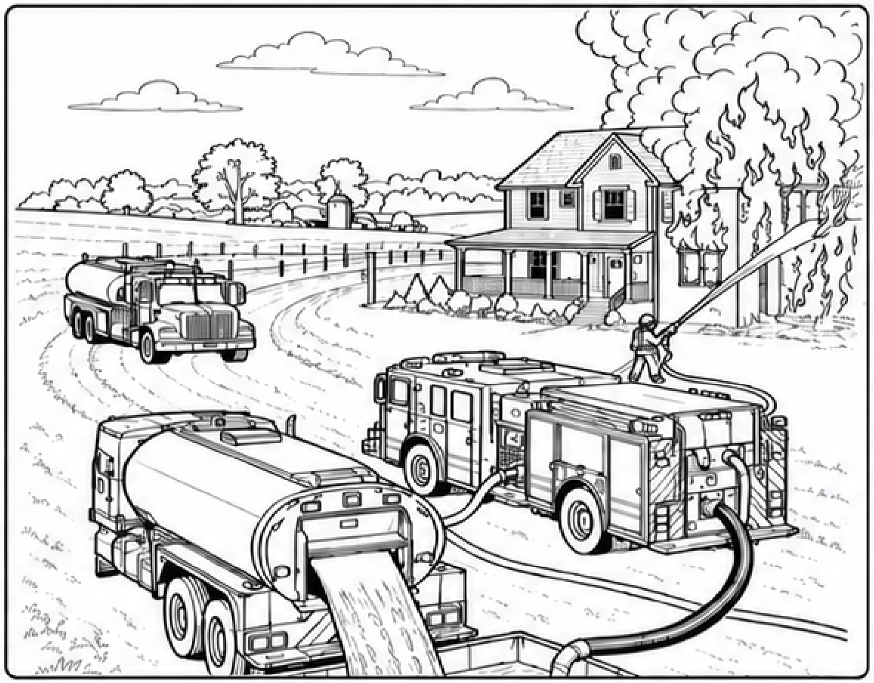
- $FL = C \times Q^2 \times L$
- $Q = GPM / 100$
- PDP = NP + FL + Appliance Loss ± Elevation
- Total Flow = Sum of All Operating Line Flows
- Water Duration (min) = Gallons Available / Total GPM
- Remaining Water = Starting Gallons – (Total GPM × Minutes)

FirePumpSim Answer Key

Driver/Operator Practice Worksheet

Student Name: ____________________ **Date:** ______________ **Class:** ______________ **Score:** __________

Scenario: Rural Water Supply - Tanker Shuttle House Fire



Given / Reference

- **Apparatus:** Engine 181 drafting from portable tank
- **Water source:** Portable dump tank supplied by water tender shuttle
- **Portable tank starting water:** 2,000 gallons
- **Attack line:** 200 ft of 1¾-inch hose
- **Nozzle:** Fog nozzle, 150 GPM at 100 PSI
- **Hose coefficient (C):** 15.5
- **Appliance loss:** 0 PSI
- **Elevation change:** 0 PSI

Instructor Answer Key

1. Friction loss in the 200-foot 1¾-inch attack line

Formula: $FL = C \times Q^2 \times L$
$Q = 150 / 100 = 1.5$
$L = 200 / 100 = 2$
$FL = 15.5 \times (1.5)^2 \times 2$
$FL = 15.5 \times 2.25 \times 2$
$FL = 69.75$ PSI
Rounded Answer: 70 PSI

2. Required pump discharge pressure (PDP)

Formula: PDP = NP + FL + Appliance Loss ± Elevation

PDP = 100 + 70 + 0 + 0

Answer: 170 PSI

3. Required PDP for the second 150-foot 2½-inch line

Formula: $FL = C \times Q^2 \times L$
$Q = 200 / 100 = 2$
$L = 150 / 100 = 1.5$
$FL = 2 \times (2)^2 \times 1.5 = 12$ PSI
PDP = 100 + 12 + 0 + 0
Answer: 112 PSI

4. Water available from the 2,000-gallon portable tank with one line flowing

Formula: Water Duration = Gallons Available / Total GPM
Water Duration = 2,000 / 150
Water Duration = 13.33 minutes
Rounded Answer: 13.3 minutes (about 13 min 20 sec)

5. Portable tank water remaining after 5 minutes with both lines flowing

Total GPM = 150 + 200 = 350 GPM
Water Used = Total GPM × Minutes
Water Used = 350 × 5 = 1,750 gallons
Remaining Water = 2,000 − 1,750 = 250 gallons
Answer: 250 gallons remaining

Accept reasonable rounding: ±1 PSI and ±0.1 minute.